

## **01. Introduction**

Cancer treatments such as surgery, chemotherapy, radiation therapy, immunotherapy, targeted therapy, and hormone therapy can cause side effects. These occur because some treatments also affect healthy cells while destroying cancer cells.

Most side effects are temporary and can be managed with proper medical care, nutrition, and lifestyle changes.

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## **02. Common Side Effects of Cancer Treatment**

Patients may experience one or more of the following:

- **Fatigue**
- **Pain**
- **Hair loss**
- **Nausea and vomiting**
- **Loss of appetite**
- **Mouth sores**
- **Diarrhea**
- **Constipation**
- **Skin changes**
- **Increased risk of infection**
- **Low blood counts**
- **Emotional stress**
- **Difficulty sleeping**

The severity varies depending on the type of treatment and the individual's overall health.

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## **03. Fatigue**

Fatigue is one of the most common side effects experienced by cancer patients.

### **Symptoms**

- **Extreme tiredness**
- **Lack of energy**
- **Weakness**
- **Difficulty concentrating**
- **Reduced physical activity**

## Management

- Get adequate rest.
  - Maintain a regular sleep schedule.
  - Perform light exercise such as walking.
  - Eat nutritious meals.
  - Stay hydrated.
  - Ask for help with daily activities.
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## 04. Nausea and Vomiting

Chemotherapy and radiation therapy may cause nausea and vomiting.

### Management

- Take anti-nausea medications as prescribed.
  - Eat small, frequent meals.
  - Avoid spicy and oily foods.
  - Drink clear fluids.
  - Eat dry crackers before getting out of bed if nausea occurs in the morning.
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## 05. Hair Loss

Hair loss occurs because chemotherapy affects rapidly dividing hair follicle cells.

Hair loss may involve:

- Scalp hair
- Eyebrows
- Eyelashes
- Body hair

### Management

- Use mild shampoo.
  - Protect the scalp from sunlight.
  - Wear hats or scarves if desired.
  - Hair usually grows back after treatment.
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## 06. Mouth Sores

**Cancer treatment may damage the lining of the mouth.**

**Symptoms**

- Pain
- Difficulty swallowing
- Dry mouth
- Ulcers

**Management**

- Brush teeth gently.
  - Use alcohol-free mouthwash.
  - Drink plenty of water.
  - Avoid spicy and acidic foods.
  - Follow oral care instructions from healthcare providers.
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**07. Loss of Appetite**

**Many patients lose interest in eating during treatment.**

**Management**

- Eat smaller meals more frequently.
  - Choose high-protein foods.
  - Drink nutritional supplements if recommended.
  - Eat favorite healthy foods.
  - Stay physically active to improve appetite.
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**08. Diarrhea**

**Certain chemotherapy drugs and radiation therapy may cause diarrhea.**

**Management**

- Drink plenty of fluids.
- Eat low-fiber foods temporarily.
- Avoid spicy foods.
- Take prescribed anti-diarrheal medication.

**Seek medical care if diarrhea is severe or persistent.**

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## **09. Constipation**

**Pain medications and reduced physical activity can cause constipation.**

### **Management**

- **Drink sufficient water.**
  - **Eat fruits and vegetables.**
  - **Include whole grains in the diet.**
  - **Walk daily if possible.**
  - **Use stool softeners only when prescribed.**
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## **10. Low Blood Counts**

**Cancer treatment can reduce blood cell production in the bone marrow.**

### **Low Red Blood Cells (Anemia)**

#### **Symptoms:**

- **Fatigue**
- **Weakness**
- **Shortness of breath**

### **Low White Blood Cells**

#### **Symptoms:**

- **Increased risk of infection**
- **Fever**
- **Chills**

### **Low Platelets**

#### **Symptoms:**

- **Easy bruising**
- **Bleeding gums**
- **Nosebleeds**

### **Management**

- **Attend regular blood tests.**
- **Follow infection prevention measures.**
- **Avoid injuries.**
- **Report fever immediately.**

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## **11. Risk of Infection**

**Chemotherapy weakens the immune system.**

### **Prevention**

- **Wash hands frequently.**
  - **Avoid crowded places when blood counts are low.**
  - **Cook food thoroughly.**
  - **Avoid contact with sick individuals.**
  - **Maintain personal hygiene.**
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## **12. Skin Changes**

**Radiation therapy may cause:**

- **Redness**
- **Dryness**
- **Itching**
- **Peeling**

### **Management**

- **Use mild soap.**
  - **Apply recommended moisturizers.**
  - **Protect skin from sunlight.**
  - **Avoid scratching affected areas.**
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## **13. Pain Management**

**Pain may result from cancer itself or its treatment.**

**Pain can usually be controlled using:**

- **Pain-relieving medicines**
- **Physical therapy**
- **Relaxation techniques**
- **Heat or cold therapy**
- **Palliative care services**

**Patients should report pain early for effective treatment.**

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## **14. Emotional and Mental Health**

**Cancer treatment often causes emotional stress.**

**Patients may experience:**

- **Anxiety**
- **Depression**
- **Fear**
- **Anger**
- **Mood changes**

**Management**

- **Counseling**
- **Support groups**
- **Family support**
- **Relaxation exercises**
- **Meditation**
- **Professional mental health care when needed**

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## **15. Sleep Problems**

**Difficulty sleeping is common.**

**Management**

- **Maintain a regular bedtime.**
- **Avoid caffeine before sleep.**
- **Limit daytime naps.**
- **Practice relaxation exercises.**
- **Keep the bedroom quiet and comfortable.**

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## **16. Importance of Nutrition**

**Good nutrition helps patients:**

- **Maintain body weight**
- **Recover faster**
- **Improve immunity**

- Reduce treatment complications
- Increase energy levels

Patients should consult a dietitian if eating becomes difficult.

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## 17. Physical Activity

Light physical activity during treatment can:

- Reduce fatigue
- Improve mood
- Maintain muscle strength
- Improve sleep quality
- Enhance overall well-being

Exercise should always be performed according to medical advice.

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## 18. When to Contact a Doctor

Patients should immediately seek medical attention if they experience:

- High fever
- Severe bleeding
- Difficulty breathing
- Persistent vomiting
- Severe diarrhea
- Chest pain
- Confusion
- Sudden swelling
- Uncontrolled pain

Early medical intervention prevents serious complications.

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## 19. Key Points

- Side effects vary depending on the type of cancer treatment.
- Most side effects can be effectively managed.
- Proper nutrition and hydration support recovery.
- Good hygiene reduces infection risk.

- Emotional health is as important as physical health.
- Patients should promptly report severe or unusual symptoms.
- Regular follow-up helps monitor and manage treatment-related complications.